

• Row-reduced Echelon form:

$$EA = R$$

• Vector space: $\mathbb{R}^n, \mathbb{F}^n$

• Basis: $\{u_1, \dots, u_n\} \subseteq V$

$\downarrow$
l.i. + Span

$$\text{Span}\{x_1, \dots, x_n\}$$

$$= \left\{ \sum_{i=1}^n \alpha_i x_i : \alpha_i \in \mathbb{F} \right\}$$

• V - f.d. if $\exists$ a basis consists finitely many vectors

• $\text{Span}\{x_1, \dots, x_m\} = V$, any linearly independent set of vectors $\{y_1, \dots, y_n\}$ satisfy $n \leq m$

• $\{x_1, \dots, x_n\}, \{y_1, \dots, y_m\}$ basis of V .
 $n \leq n$ and $n \leq m \Rightarrow n = m$

V - f.d.

$$W \subseteq V$$

$$\{w_1, \dots, w_n\}, (\neq) x \in V \setminus W$$

$$\Rightarrow \{x, w_1, \dots, w_n\} \text{ is also l.i.}$$

$$V, \{u_1, u_2, \dots, u_n\}$$

$$\text{Span}\{u_1\} \neq V, u_2 \in V \setminus \text{Span}\{u_1\}$$

$$Y \subseteq X, X/Y$$

$$\dim(X) = \dim(Y) + \dim(X/Y)$$

$$\dim(W_1 \oplus W_2) = \dim(W_1) + \dim(W_2)$$

Linear maps: $T: V \rightarrow W$

$$T(\alpha v_1 + \beta v_2) = \alpha T(v_1) + \beta T(v_2)$$

Isomorphism

$$T: V \xrightarrow{\text{iso}} W$$

$$\begin{array}{ccc} \text{basis for } V & \leftarrow \{v_1, \dots, v_n\} & \rightarrow \{T(v_1), \dots, T(v_n)\} \\ & & \downarrow \\ & & \text{basis for } W \end{array}$$

$$\dim(V) = \dim(W) \Leftrightarrow V \cong W$$

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \{v_1, \dots, v_n\} & & \{w_1, \dots, w_n\} \end{array}$$

$$T(v_i) = w_i, \forall i=1, \dots, n$$

$$\mathbb{R}^n \xrightarrow{T} \mathbb{R}^m$$

$$A = \begin{bmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{bmatrix} : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

($e_1, \dots, e_n$) basis

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}_{n \times 1} \rightarrow \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}_{m \times 1}$$

$$A(e_i) = T(e_i), \quad A(e_i) = T(e_i) \quad \forall i=1, \dots, n$$

$$\tilde{T} = \begin{bmatrix} A_{11} \\ A_{21} \\ \vdots \\ A_{m1} \end{bmatrix}_{m \times n}$$

$$T(v_i) = \sum_{j=1}^m A_{ji} w_j$$

$$V \xrightarrow{T} W = \{w_1, \dots, w_m\}$$

$$\begin{array}{ccc} \mathbb{R}^n & \xrightarrow{\tilde{T}} & \mathbb{R}^m \\ (e_1, \dots, e_n) & & (e_1, \dots, e_m) \end{array}$$

$$D(a_1, \dots, a_n) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1)} \dots a_{\sigma(n)}$$

$$\det A = \det(a_1, \dots, a_n) := D(a_1, \dots, a_n)$$

$$\det(A B) = \det(A) \det(B)$$

$$\text{tr} : M_{n \times n}(\mathbb{R}) \longrightarrow \mathbb{R}$$

$$\text{tr}(AB) = \text{tr}(BA)$$

$$A \in M_{n \times n}(\mathbb{R})$$

$$x \in \sigma(A) \Rightarrow$$

$$\exists \lambda(\neq 0) \\ Ax = \lambda x,$$

$$D : \underbrace{\mathbb{R}^n \times \dots \times \mathbb{R}^n}_{n \text{ times}} \longrightarrow \mathbb{R}$$

$$(i) D(e_1, \dots, e_n) = 1$$

$$(ii) D(\lambda_1, \dots, \lambda_n) = 0, \text{ if } \lambda_i = \lambda_j \text{ (i \neq j)}$$